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Charging control system, charging control apparatus, and computer readable recording medium

Abstract

A charging control system includes: a charging apparatus including a first processor, the charging apparatus being configured to store electric power to be supplied in a predetermined area; a charging stand provided in the predetermined area and including a second processor, the charging stand being configured to supply the electric power supplied from the charging apparatus to one or more electric vehicles in the predetermined area; and a charging control apparatus including a third processor, the third processor being configured to calculate number of the one or more electric vehicles in the predetermined area, calculate a charging amount of the charging apparatus based on the calculated number of the one or more electric vehicles, and perform charging control of the charging apparatus based on the calculated charging amount.

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Background/Summary

(1) The present application claims priority to and incorporates by reference the entire contents of Japanese Patent Application No. 2020-194668 filed in Japan on Nov. 24, 2020.

BACKGROUND

(2) The present disclosure relates to a charging control system, a charging control apparatus, and a computer readable recording medium.

(3) There is a known technique for supplying electric power stored in a charging apparatus in advance to electric vehicles (for example, a battery electric vehicle, a plug-in hybrid electric vehicle, and the like) positioned in a smart city.

SUMMARY

(4) In the smart city, a charging stand for charging the electric vehicles entering and leaving the smart city is provided.

(5) There is a need for a charging control system, a charging control apparatus, and a computer readable recording medium that are able to handle charging requests from a large number of electric vehicles entering and leaving a smart city.

(6) According to one aspect of the present disclosure, there is provided a charging control system including: a charging apparatus including a first processor, the charging apparatus being configured to store electric power to be supplied in a predetermined area; a charging stand provided in the predetermined area and including a second processor, the charging stand being configured to supply the electric power supplied from the charging apparatus to one or more electric vehicles in the predetermined area; and a charging control apparatus including a third processor, the third processor being configured to calculate number of the one or more electric vehicles in the predetermined area, calculate a charging amount of the charging apparatus based on the calculated number of the one or more electric vehicles, and perform charging control of the charging apparatus based on the calculated charging amount.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a block diagram illustrating a charging control system according to an embodiment; and

(2) FIG. 2 is a flowchart illustrating an example of a charging control method performed by the charging control system according to the embodiment.

DETAILED DESCRIPTION

(3) A charging control system, a charging control apparatus, and a computer readable recording medium storing a charging control program according to an embodiment of the present disclosure will be described with reference to the drawings. Note that, the constituent elements in the following embodiment include those that may be easily replaced by those skilled in the art or are

substantially the same.

(4) A charging control system including a charging control apparatus according to the embodiment will be described with reference to FIG. 1. The charging control system performs charging control of a charging apparatus that supplies electric power to a preset area. The “preset area” is an area where a large number of users live or that a large number of users use and is, for example, a smart city, a theme park, an amusement park, or the like. The present embodiment is described on the assumption that the area is a smart city.

(5) As illustrated in FIG. 1, a charging control system **1** includes a charging control apparatus **10**, a plurality of electric vehicles **20**, a plurality of charging apparatuses **30**, and an entrance/exit gate **50**. The charging control apparatus **10**, the electric vehicles **20**, the charging apparatuses **30**, and the entrance/exit gate **50** each have a communication function and are configured to be communicable with each other via a network NW. This network NW is implemented by, for example, an Internet network, a mobile phone network, or the like.

(6) The charging control apparatus **10** is provided inside the smart city or outside the smart city. The charging control apparatus **10** is implemented by a general-purpose computer, such as a workstation or a personal computer.

(7) As illustrated in FIG. 1, the charging control apparatus **10** includes a control unit **11**, a communication unit **12**, and a storage unit **13**. The control unit **11** specifically includes a processor implemented by a central processing unit (CPU), a digital signal processor (DSP), a field-programmable gate array (FPGA) or the like, and a memory (main storage unit) implemented by a random access memory (RAM), a read only memory (ROM), or the like.

(8) The control unit **11** loads and executes a program stored in the storage unit **13** in the work area of the main storage unit and controls each component or the like through the execution of the program to implement a function matching a predetermined purpose. The control unit **11** functions as a number-of-vehicles calculation unit **111**, a charging amount calculation unit **112**, and a charging control unit **113** through the execution of the program stored in the storage unit **13**.

(9) The number-of-vehicles calculation unit **111** calculates the number of the electric vehicles **20** in a preset area (hereinafter, referred to as a “smart city”). A specific method of calculating the number of the electric vehicles **20** by the number-of-vehicles calculation unit **111** is not particularly limited, and various prediction methods may be used.

(10) For example, the number-of-vehicles calculation unit **111** may calculate the number of the electric vehicles **20** in the smart city based on a future scheduled travel route of each electric vehicle **20**. In this case, the vehicle number-of-vehicles calculation unit **111** acquires, from each electric vehicle **20**, information regarding a destination or a waypoint of the electric vehicle **20** (hereinafter, referred to as “traveling schedule information”) via the network NW. Then, the number-of-vehicles calculation unit **111** refers to the traveling schedule information and calculates, as the number of the electric vehicles **20** in the smart city, the number of the electric vehicles **20** setting the smart city as the destination or the waypoint.

(11) In addition, the number-of-vehicles calculation unit **111** may calculate the number of the electric vehicles **20** in the smart city based on position information about each electric vehicle **20**. In this case, the vehicle number-of-vehicles calculation unit **111** acquires, from each electric vehicle **20**, information regarding the current position of the electric vehicle **20** (hereinafter, referred to as “vehicle position information”) via the network NW. Then, the vehicle number-of-vehicles calculation unit **111** calculates the number of the electric vehicles **20** in the smart city based on the acquired vehicle position information. Note that, the vehicle position information is acquired by, for example, a GPS receiver mounted on each electric vehicle **20**. In this manner, by using the position information about each electric vehicle **20**, it is possible to easily calculate the number of the electric vehicles **20** entering and leaving the smart city.

(12) In addition, the number-of-vehicles calculation unit **111** may actually count the number of the electric vehicles **20** entering and leaving the smart city. In this case, the number-of-vehicles

calculation unit **111** acquires an image of the electric vehicles **20** passing through the entrance/exit gate **50** from a camera **51** provided at the entrance/exit gate **50**. Then, the vehicle number-of-vehicles calculation unit **111** performs image processing on the image to discriminate the electric vehicles **20** and calculates the number of the electric vehicles **20** in the smart city based on the discrimination result. The entrance/exit gate **50** is provided at the entrance of the smart city. In this manner, by using the image by the camera **51** provided at the entrance/exit gate **50**, it is possible to more accurately calculate the number of the electric vehicles **20** entering and leaving the smart city. (13) In addition, the number-of-vehicles calculation unit **111** may calculate only the number of the electric vehicles **20** that need to be charged among the electric vehicles **20** entering and leaving the smart city. In this case, the number-of-vehicles calculation unit **111** calculates the number of the electric vehicles **20** in the smart city by using any one of (1) a method using the future scheduled travel routes of the electric vehicles **20**, (2) a method using the position information about the electric vehicles **20**, and (3) a method of actually counting the number of the electric vehicles **20** entering and leaving the smart city. Subsequently, the number-of-vehicles calculation unit **111** acquires, from each electric vehicle **20** in the smart city, information regarding the remaining electric power of the electric vehicle **20** (hereinafter, referred to as “SOC information”) via the network NW. Then, the number-of-vehicles calculation unit **111** refers to the SOC information and calculates the number of the electric vehicles **20** whose remaining electric power is equal to or less than a preset value. In this manner, by calculating only the number of the electric vehicles **20** that need to be charged, it is possible to prevent excessive electric power from being stored in the charging apparatuses **30**.

(14) The charging amount calculation unit **112** calculates the charging amount of the charging apparatuses **30** based on the number of the electric vehicles **20** calculated by the number-of-vehicles calculation unit **111**. The charging amount calculation unit **112** calculates the charging amount of the charging apparatuses **30** by, for example, multiplying a preset charging amount for one electric vehicle **20** by the number of the electric vehicles **20** calculated by the number-of-vehicles calculation unit **111**.

(15) In addition, the charging amount calculation unit **112** may calculate the charging amount of the charging apparatuses **30** in consideration of the remaining electric power of each electric vehicle **20** in addition to the number of the electric vehicles **20**. In this case, the charging amount calculation unit **112** acquires the SOC information from each electric vehicle **20** in the smart city via the network NW. Then, the charging amount calculation unit **112** calculates the charging amount of the charging apparatuses **30** based on the number of the electric vehicles **20** calculated by the number-of-vehicles calculation unit **111** and the SOC information. In this manner, by calculating the charging amount in consideration of the remaining electric power of each electric vehicle **20**, it is possible to prevent excessive electric power from being stored in the charging apparatuses **30**.

(16) The charging control unit **113** performs charging control of the charging apparatuses **30** based on the charging amount calculated by the charging amount calculation unit **112**. The charging control unit **113** performs charging control by storing, in the charging apparatuses **30**, the electric power corresponding to the charging amount calculated by the charging amount calculation unit **112**. A specific method of the charging control by the charging control unit **113** is not particularly limited, and the charging control may be performed by, for example, determining a final charging amount based on the charging amount calculated by the charging amount calculation unit **112** and on charging information (for example, a current charging capacity or the like) acquired in advance from the charging apparatuses **30** and transmitting, to each charging apparatus **30**, a command to increase or decrease a current or future charging amount according to the final charging amount.

(17) The communication unit **12** is implemented by, for example, a local area network (LAN) interface board, a wireless communication circuit for wireless communication, or the like. The communication unit **12** is connected to the network NW such as the Internet, which is a public communication network. Then, the communication unit **12** connects to the network NW to

communicate with the electric vehicles **20**, the charging apparatuses **30**, and the entrance/exit gate **50**.

(18) The storage unit **13** is implemented by recording media, such as an erasable programmable ROM (EPROM), a hard disk drive (HDD), a removable medium, and the like. Example of the removable media include disc recording media, such as a universal serial bus (USB) memory, a compact disc (CD), a digital versatile disc (DVD), and a Blu-ray (registered trademark) disc (BD). The storage unit **13** may store an operating system (OS), various programs, various tables, various databases, and the like. The storage unit **13** further stores, for example, the number of the electric vehicles **20** calculated by the number-of-vehicles calculation unit **111**, the charging amount calculated by the charging amount calculation unit **112**, the charging amount of the charging apparatuses **30** determined by the charging control unit **113**, and the like as necessary.

(19) The electric vehicles **20** each are a moving body capable of communicating with the outside. The electric vehicles **20** each are implemented by a battery electric vehicle or a plug-in hybrid electric vehicle that may be charged from an external power supply. The electric vehicles **20** each transmit the vehicle position information and the SOC information to the charging control apparatus **10** sequentially or at a predetermined cycle as necessary.

(20) The charging apparatuses **30** each store electric power to be supplied to the equipment in the smart city and are provided inside the smart city or outside the smart city. The charging apparatuses **30** are each connected to a charging stand **40** provided in the smart city and are configured to be able to supply electric power to the charging stand **40**.

(21) The charging apparatuses **30** are each implemented by a general-purpose computer such as a workstation, a personal computer, and the like. In addition, the charging apparatuses **30** each include a battery for storing electric power, a control mechanism for controlling charging and discharging of electric power, and the like. In addition, the charging apparatuses **30** each transmit current charging information regarding each charging apparatus **30** to the charging control apparatus **10** sequentially or at a predetermined cycle.

(22) The charging stand **40** supplies electric power to the electric vehicles **20** in the smart city and is implemented by, for example, a contact type charging stand, a non-contact type charging stand, or the like.

(23) The entrance/exit gate **50** is provided at the entrance of the smart city. The entrance/exit gate **50** is provided with a plurality of cameras **51** that photographs the electric vehicles **20** passing through the entrance/exit gate **50**. The entrance/exit gate **50** transmits the images photographed by the cameras **51** to the charging control apparatus **10** sequentially or at a predetermined cycle.

(24) An example of a processing procedure of a charging control method performed by the charging control system **1** according to the embodiment will be described with reference to FIG. 2. Note that, the following is a description of an example of calculating the number of the electric vehicles **20** in the smart city based on future scheduled travel routes of the electric vehicles **20**.

(25) First, each charging apparatus **30** transmits charging information to the charging control apparatus **10** (step S1). Then, each electric vehicle **20** transmits traveling schedule information to the charging control apparatus **10** (step S2). Note that, the order of steps S1 and S2 may be reversed.

(26) Then, the number-of-vehicles calculation unit **111** of the charging control apparatus **10** refers to the traveling schedule information and calculates, as the number of the electric vehicles **20** in the smart city, the number of the electric vehicles **20** setting the smart city as a destination or a waypoint (step S3). Then, the charging amount calculation unit **112** calculates a charging amount based on the number of the electric vehicles **20** calculated in step S3 (step S4). Then, the charging control unit **113** performs charging control of the charging apparatuses **30** based on the charging amount calculated in step S4 (step S5). With the above steps, the processing of the charging control method is completed.

(27) As described above, with the charging control system, the charging control apparatus, and the

charging control program according to the embodiment, it is possible to handle charging requests from a large number of the electric vehicles **20** entering and leaving the smart city.

(28) That is, the charging control system, the charging control apparatus, and the charging control program according to the embodiment perform the charging control of the charging apparatuses **30** in consideration of the number of the electric vehicles **20** entering and leaving the smart city.

Accordingly, if there are a large number of the electric vehicles **20**, electric power to be supplied to the equipment and facilities in the smart city does not run short.

(29) In addition, the charging control system, the charging control apparatus, and the charging control program according to the embodiment do not simply predict an electric power demand in the smart city but predict (calculate) the number of the electric vehicles **20** entering and leaving the smart city and take the result into consideration. Accordingly, it is possible to more appropriately perform the charging control of the charging apparatuses **30**.

(30) According to the present disclosure, it is possible to handle charging requests from a large number of electric vehicles entering and leaving a smart city.

(31) Although the disclosure has been described with respect to specific embodiments for a complete and clear disclosure, the appended claims are not to be thus limited but are to be construed as embodying all modifications and alternative constructions that may occur to one skilled in the art that fairly fall within the basic teaching herein set forth.

Claims

1. A charging control apparatus comprising a processor configured to: calculate number of one or more electric vehicles in a preset area; calculate, based on the calculated number of the one or more electric vehicles, a charging amount of a charging apparatus configured to supply electric power to the one or more electric vehicles through a charging stand provided in the preset area; perform charging control of the charging apparatus based on the calculated charging amount; acquire information about a destination or a waypoint of each of the one or more electric vehicles traveling both within and outside the preset area; and calculate, as the number of the one or more electric vehicles in the preset area, the number of the one or more electric vehicles setting the preset area as the destination or the waypoint.

2. The charging control apparatus according to claim 1, wherein the processor is configured to: acquire information regarding remaining electric power of each of the one or more electric vehicles in the preset area; and calculate a charging amount of the charging apparatus based on the calculated number of the one or more electric vehicles and the information regarding the remaining electric power of each of the one or more electric vehicles.
